

Amrita School of Computing, Amritapuri
Department of Computer Science and Engineering(AI)
B.Tech 2022-2026 CSE(AI)
Project Phase -Zeroth Proposal

1. Project Group No/ID: 5

2. Team Members:

Reg No	Name
AM.EN.U4AIE22002	Abhiram A S
AM.EN.U4AIE22028	Karthik Narayan C
AM.EN.U4AIE22119	Harigovind C B

3. Proposed Project Title:

A Framework for the Syntactic and Semantic Analysis of Ayurvedic Texts using Artificial Intelligence.

4. Give a brief description of the project:

This project applies Large Language Models (LLMs) and NLP techniques to process classical Ayurvedic texts written in Sanskrit, such as the *Charaka Samhitā* and *Suśruta Samhitā*.

The aim is to perform syntactic analysis (sandhi splitting, morphological tagging, dependency parsing) and semantic analysis (NER, relation extraction) to identify entities like medicinal herbs, diseases, and their relationships.

Extracted knowledge will be structured into a queryable Knowledge Graph (KG) using Neo4j.

This system will enable semantic search and advanced querying of Ayurvedic medical knowledge.

The project addresses challenges of Sanskrit's morphological richness, low-resource status, and domain-specific entity recognition.

5. Had an initial discussion with the guide: Yes

6. Identified at least 3 (journal/conference/other sources) after discussion with the guide: Yes

7. Give the Bibliographic details of resources identified after confirmation from the guide:(No need for description, only list 3 papers)

Title	Source	Year of Publication
SanskritShala: A Neural Sanskrit NLP Toolkit with Web-Based Interface	ACL System Demonstrations	2023
Semantic Annotation and Querying Framework Based on Semi-structured Ayurvedic Text	arXiv	2022
HerbKG: Constructing a Herbal–Molecular Medicine Knowledge Graph	Frontiers in Genetics	2022

8. Software / Hardware Requirements

Software: Python, SanskritShala toolkit, Samsaadhanii Sandhi Splitter, Hugging Face Transformers, Neo4j, PyTorch, spaCy, NLTK, pyvis.

Hardware: Standard development laptop, optional GPU access for model fine-tuning.

9. Whether Software/ Hardware is available?

Yes

10. If No, what is the alternate plan?

Biomedical Natural Language Processing (Biomedical NLP)

Students' Name and Signature

Abhiram A S	
Karthik Narayan C	
Harigovind C B	

Guide's Name and Signature

Jayasree Nair

Date

11-08-2025